



PAKISTAN RAILWAYS

UI / UX Redesign Project

HCI & Computer Graphics Course | May 2026

Built with: HTML5 | CSS3 | Bootstrap 5 | Vanilla JavaScript

Presentation Agenda

01 Project Overview & Problem Statement

02 Analysis of Original Website

03 HCI Principles Implemented

04 Design System & Color Palette

05 Website Pages & Features

06 Technical Implementation

07 Accessibility Audit

08 Usability Improvements

09 Live Demo Walkthrough

10 Conclusion & Future Work

Project Overview

Pakistan Railways

National railway of Pakistan — est. 1861
7,791 km of track across the country
450+ stations, 500+ daily trains
65,000 employees serving the nation
Critical public transport infrastructure

Our Mission

Redesign the Pakistan Railways website to create a modern, accessible, and user-friendly portal that genuinely serves Pakistan's rail passengers using HTML, CSS, and JavaScript.

7,791 km
Track Network

450+
Stations

500+
Daily Trains

65,000+
Employees

Problems with Original pakrail.gov.pk



Not Mobile Responsive

Completely unusable on smartphones and tablets — no adaptive layout



Outdated Visual Design

Low contrast, inconsistent colors, poor typography, dated aesthetics



Confusing Navigation

Poorly labelled menus, no active states, nested links hard to find



No User Feedback

Forms give no validation, no loading indicators, no confirmation



Inaccessible

No ARIA labels, no keyboard support, no skip navigation, fails WCAG



Hard to Search

No autocomplete, exact station names required, no real-time results

5 HCI Principles Implemented

Every design decision was guided by these core Human-Computer Interaction principles



ACCESSIBILITY

WCAG 2.1 AA compliant
ARIA labels, keyboard nav
Skip links, focus indicators



CONSISTENCY

Unified design system
CSS variables, same
components
across all 6 pages



NAVIGATION

Sticky navbar, breadcrumbs
Active states, Back-to-top
Mobile hamburger menu



USER FEEDBACK

Toast notifications
Real-time validation
Progress indicators



EFFICIENCY

Autocomplete, recent
searches
Quick routes, URL params
Keyboard shortcuts

Accessibility

WCAG 2.1 Level AA Compliant



Skip Navigation

Skip to main content link at top of every page



Keyboard Support

Full Tab/Enter/Space/Esc keyboard navigation



Color Contrast

7.2:1 ratio — exceeds 4.5:1 AA minimum



Form Labels

Every input has associated <label> element



ARIA Labels

All buttons, inputs, and nav elements are labeled



Focus Indicators

3px gold focus rings on all interactive elements



Semantic HTML

<nav>, <main>, <section>, <article> used correctly



Live Regions

aria-live="polite" on dynamic toast notifications



Consistency & Design System

Color Palette



Primary Green
#006633



Dark Green
#004D26



PR Gold
#E8A020



Dark Navy
#1A1A2E

Typography

Poppins (Headings)

Inter — clean, readable body font for long-form content

Consistent Components

20+ CSS variables for theming (colors, spacing, radii)

Identical header/footer across all 6 pages

Unified button system: Primary, Gold, Outline variants

Feature cards, route cards, stat bars — reused throughout

Same form styles, same validation messages

Consistent 8px spacing grid

Navigation Features

- Sticky navbar — always visible while scrolling
- Active page indicator — gold underline on current page
- Breadcrumbs on all inner pages
- Mobile hamburger menu (Bootstrap collapse)
- Back to top button after 400px scroll
- "Book Ticket" CTA always in gold in navbar
- Footer site map with all page links
- "/" keyboard shortcut focuses search

User Feedback Features

- Custom toast notifications (success/error/warning)
- Real-time form validation with green/red indicators
- Specific helpful error messages per field
- Button loading spinner during payment processing
- Full booking confirmation with reference number
- Status badges: On Time / Delayed / Cancelled
- Journey progress bar with percentage
- Counter animation on statistics (0→target)

⚡ Efficiency — Smart Interactions



Station Autocomplete

36 stations in datalist — start typing for suggestions on all route inputs



Recent Searches

5 most recent searches saved in localStorage and shown as quick-access pills



Popular Routes

6 pre-configured routes on homepage — click to auto-fill search form



Station Swap

One-click button to swap origin and destination instantly



Smart Date Defaults

Date inputs default to today, prevent past date selection automatically



CNIC Auto-Format

CNIC input auto-formats as user types: XXXXX-XXXXXXX-X pattern



Card Formatting

Credit card auto-inserts spaces every 4 digits for easy reading



URL Parameters

From homepage to schedule page — search params passed via URL automatically

6-Page Website Overview



Home

`index.html`

Hero search, popular routes, booking steps, stats, news



Schedule

`schedule.html`

Search 10 trains, sort by price/time/duration, book directly



Track Train

`track.html`

Live timeline for 8 trains with animated progress bar



Booking

`booking.html`

4-step flow: Journey → Passenger → Payment → Confirmation



About

`about.html`

History since 1861, fleet classes, concession fares



Contact

`contact.html`

Form, 4 regional offices, FAQ accordion, helpline 117

Technical Implementation

Built with pure HTML, CSS, and JavaScript — no frameworks

HTML5

Semantic markup
<nav>, <main>,
<section>
ARIA attributes
Form elements

CSS3

Custom Properties
Flexbox & Grid
Animations & transitions
Media queries

Bootstrap 5

Responsive grid
Navbar collapse
Accordion & tabs
Utility classes

JavaScript

ES6+ Vanilla JS
DOM manipulation
LocalStorage API
IntersectionObserver

Font Awesome

100+ icons
Consistent iconography
Accessibility icons
Social media

Before vs After — Usability Comparison

FEATURE	ORIGINAL SITE	REDESIGNED SITE
Mobile Responsive	✗ Broken on phones	✓ Mobile-first design
Train Schedule	✗ Static hard-coded table	✓ Searchable + sortable
Ticket Booking	✗ External PDF links	✓ Full 4-step online flow
Train Tracking	✗ Not available	✓ Live timeline visualization
Form Validation	✗ No feedback at all	✓ Real-time field validation
Accessibility	✗ Fails WCAG checks	✓ WCAG 2.1 AA compliant
Navigation	✗ Confusing menus	✓ Sticky, active, breadcrumbs
Search	✗ Manual exact text	✓ Autocomplete + recent searches

Accessibility Audit — WCAG 2.1 Results

Criterion	Description	Level	Status	Notes
1.1.1	Non-text Content	A	PASS	All icons have aria-hidden; images have alt text
1.3.1	Info and Relationships	A	PASS	Semantic HTML: nav, main, section, header, footer
1.4.3	Contrast Minimum	AA	PASS	7.2:1 achieved — exceeds 4.5:1 requirement
2.1.1	Keyboard Accessible	A	PASS	All functionality accessible via keyboard only
2.4.1	Bypass Blocks	A	PASS	Skip to main content link on every page
2.4.4	Link Purpose	A	PASS	All links have descriptive text or aria-label
3.3.1	Error Identification	A	PASS	Field-level error messages with specific guidance
3.3.2	Labels or Instructions	A	PASS	All form inputs have explicit <label> elements
4.1.2	Name, Role, Value	A	PASS	ARIA attributes on interactive controls
4.1.3	Status Messages	AA	PASS	aria-live="polite" on dynamic content regions

Conclusion

- ✓ 6 pages built with HTML/CSS/JS — no modern frameworks
- ✓ 5 HCI principles fully implemented and demonstrated
- ✓ WCAG 2.1 AA accessibility compliance achieved
- ✓ Mobile-first responsive design across all screen sizes
- ✓ Real ticket booking flow, schedule search, train tracking
- ✓ Professional government-grade portal aesthetic

"A government portal that citizens actually want to use — accessible, consistent, and efficient."



Thank You!

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Questions & Discussion